

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

ENTERPRISE INFRASTRUCTURE PLAN

for Vantix Technologies

ITEP 414 — System Administration and Maintenance

Week 2 Portfolio Project — Individual Portfolio Project

Program: Bachelor of Science in Information Technology

Prepared by:

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Role: Junior System Administrator, Vantix Technologies

Part 1 — Company Profile

Company Name: Vantix Technologies

Nature of Business: A newly established software development company that builds custom web and mobile applications, and provides IT consulting services for small and medium-sized businesses.

Company Vision: To become a trusted regional technology partner recognized for delivering reliable, secure, and innovative software solutions that help small businesses grow.

Company Mission: To design and deliver high-quality software and IT solutions through a well-planned, secure, and scalable technology infrastructure, while fostering a culture of continuous learning, professionalism, and client-focused service among our team.

Office Location (fictional): 3rd Floor, Meridian Business Tower, National Highway, Brgy. Halang, Calamba City, Laguna, Philippines 4027

Organizational Structure

Vantix Technologies operates under a flat organizational structure typical of an early-stage startup, with four functional departments reporting to a Chief Executive Officer (CEO):

- Chief Executive Officer (CEO) — overall business direction and strategy
 - IT Department — headed by the IT Manager; includes the Junior System Administrator, developers, and helpdesk support
 - Human Resources Department — headed by the HR Manager; handles recruitment, employee records, and payroll coordination
 - Finance Department — headed by the Finance Manager; handles accounting, invoicing, and financial reporting
 - Sales Department — headed by the Sales Manager; handles client acquisition, account management, and field sales

Employee Distribution

Department	Employees	% of Workforce
Information Technology	5	25%
Human Resources	4	20%
Finance	5	25%
Sales	6	30%
TOTAL	20	100%

Part 2 — Enterprise Hardware Inventory

The hardware inventory below was designed so that every one of the company's 20 employees has an appropriate primary device, while shared infrastructure (server, network equipment, printers) is sized for reliability and future growth. Department-specific choices — such as laptops for Sales and fixed desktops for Finance — are based on each department's actual workflow and data-security needs.

Asset ID	Hardware	Qty	Department	Purpose / Justification
HW-001	Desktop Computer	4	IT Department	Software development, testing, and system administration workstations
HW-002	Desktop Computer	4	Human Resources	Employee records management, payroll processing, and office productivity
HW-003	Desktop Computer	5	Finance	Accounting, invoicing, and financial reporting (fixed workstations for data security)
HW-004	Desktop Computer	1	Sales	In-office sales coordination and CRM data entry
HW-005	Laptop	1	IT Department	On-site troubleshooting, server maintenance, and remote support
HW-006	Laptop	5	Sales	Field sales, client visits, and offsite presentations
HW-007	Server (Tower/Rack)	1	IT Department	Hosts file sharing, internal applications, Git repository, and virtualization (VirtualBox)
HW-008	Router	1	IT Department (Network)	Distributes internet connection and routes traffic between LAN and WAN
HW-009	Managed Switch (24-port)	1	IT Department (Network)	Central connectivity point for all wired department devices
HW-010	Network Printer	2	Shared (HR/Finance/Sales)	Shared printing, scanning, and photocopying for daily operations
HW-011	UPS (Uninterruptible Power Supply)	3	Server Room / Network Rack	Power backup to protect server, router, switch, and firewall from outages
HW-012	Wireless Access Point	2	IT Department (Network)	Provides Wi-Fi coverage for laptops, mobile devices, and guests across the floor
HW-013	NAS Storage	1	IT Department	Centralized shared storage and automated backup repository
HW-014	External Backup Drive	2	IT Department	Offsite/rotational backup copies for disaster recovery
HW-015	Monitor (Single)	14	All Departments	One monitor per desktop workstation
HW-016	Monitor (Dual-setup, 2nd unit)	4	IT Department	Additional monitor for developers to increase coding/debugging productivity

Justification Highlights

- Sales staff receive laptops instead of desktops because their work requires client visits and offsite presentations.
- Finance staff use fixed desktop computers only (no laptops) to reduce the risk of sensitive financial data leaving the office on portable devices.
- The IT team receives dual monitors to improve productivity when coding, debugging, and monitoring systems simultaneously.
- Three UPS units protect the server, network rack, and reception/critical desktops from power interruptions common in the area.
- A NAS device plus rotating external drives support the 3-2-1 backup strategy described in Part 7.

Part 3 — Enterprise Software Inventory

The software stack below balances licensed productivity tools with free/open-source development and utility tools, keeping recurring costs low for a startup while still meeting professional development and security standards.

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume Licensing (per device)	Primary operating system for all desktop and laptop workstations; supports domain join, BitLocker, and remote management
Ubuntu Server	22.04 LTS	Open Source (Free)	Server operating system for hosting internal applications, file sharing, and the Git repository
Microsoft Office	365 Business Standard	Subscription (per user)	Word processing, spreadsheets, email, and presentations for all departments
Visual Studio Code	1.9x (latest)	Free / Open Source	Primary source-code editor for the IT/Development team
Git	2.45	Free / Open Source	Distributed version control for tracking and managing source code changes
GitHub Desktop	3.4	Free	Graphical Git client used by developers to manage repositories without the command line
VirtualBox	7.0	Free / Open Source (GPL)	Creates isolated virtual machines for testing, staging, and training environments
Google Chrome	Latest Stable	Free	Standard company-wide web browser for daily operations and web-based tools
Microsoft Defender	Built-in (Windows 11)	Included with Windows License	Real-time antivirus and anti-malware protection on all

Software	Version	License	Purpose
			endpoints
AnyDesk	8.0	Free / Business (IT use)	Remote desktop access used by the Helpdesk/IT team for troubleshooting
7-Zip	23.01	Free / Open Source	File compression and archiving for backups and file transfers

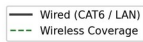
Part 4 — Enterprise Network Inventory

This inventory covers the core networking equipment required to connect all departments, provide wired and wireless access, and give the network room to grow.

Asset ID	Equipment	Qty	Purpose
NW-001	ISP Modem	1	Converts the ISP's incoming signal (fiber/DSL) into a usable Ethernet connection for the router
NW-002	Router	1	Manages traffic between the internal network (LAN) and the internet (WAN); handles NAT and DHCP
NW-003	Firewall (hardware appliance)	1	Filters inbound/outbound traffic, blocks intrusions, and enforces network security policies
NW-004	Managed Switch (24-port)	1	Provides wired connectivity and VLAN segmentation for all departments
NW-005	Wireless Access Point	2	Extends network access wirelessly for laptops and mobile devices across the office floor
NW-006	Patch Panel (24-port)	1	Organizes and terminates structured CAT6 cabling from wall outlets to the switch
NW-007	CAT6 Cable (box, 305m)	2	Structured cabling for all wired workstation and device connections
NW-008	RJ45 Connectors (box of 100)	2	Termination connectors for CAT6 cable runs and patch cables

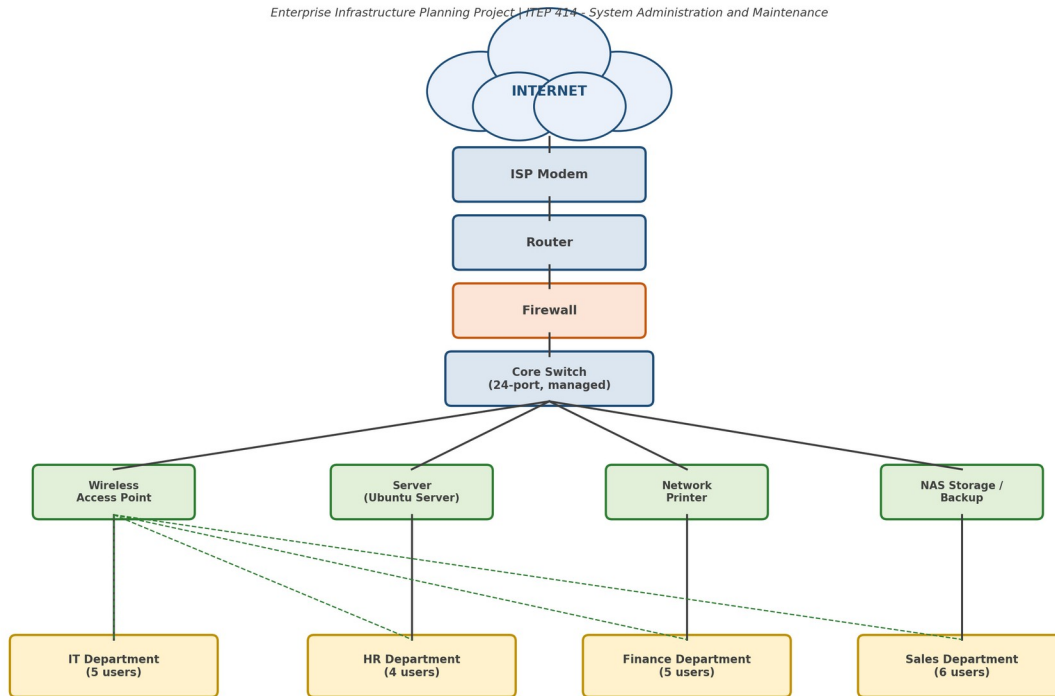
Part 5 — Enterprise Network Diagram

The diagram below shows the complete network topology for Vantix Technologies, from the internet connection down to each department's access. Traffic flows from the Internet through the ISP Modem, Router, and Firewall, into a central managed Switch, which distributes wired connectivity to the Server, Printer, and NAS storage, and both wired and wireless connectivity to the four departments through the Wireless Access Point.



Vantix Technologies — Enterprise Network Topology

Enterprise Infrastructure Planning Project | ITEP 414 - System Administration and Maintenance



Prepared by: [Your Full Name] | BSIT - System Administration and Maintenance (ITEP 414)

Week 2 Portfolio Project

Editable source file: [diagrams/network_diagram.drawio](#) (create in Draw.io using this layout). Exported copies are provided as PNG and PDF in the `diagrams/` folder of the GitHub repository.

Part 6 — System Administration Roles

Helpdesk Technician

Responsibilities:

- Provide first-line technical support to employees for hardware, software, and account issues
- Log, track, and resolve support tickets within agreed response times
- Set up new workstations, install software, and perform basic troubleshooting
- Escalate complex issues to Network or System Administrators

Skills:

- Strong communication and customer-service skills
- Basic knowledge of Windows/Office troubleshooting
- Familiarity with ticketing systems and remote-support tools
- Patience and problem-solving under pressure

Common Tools:

- AnyDesk / TeamViewer (remote support)
- Ticketing systems (e.g., Freshservice, Zendesk)
- Microsoft Defender / antivirus consoles
- Active Directory Users and Computers (basic account resets)

Certifications:

- CompTIA A+
- Microsoft Certified: Modern Desktop Administrator (legacy)
- ITIL Foundation

Network Administrator

Responsibilities:

- Design, configure, and maintain routers, switches, firewalls, and wireless access points
- Monitor network performance, uptime, and security
- Manage IP addressing, VLANs, and bandwidth allocation
- Respond to and resolve network outages and connectivity issues

Skills:

- Deep understanding of TCP/IP, routing, and switching
- Firewall configuration and network security practices
- Ability to read and design network topology diagrams
- Troubleshooting under time pressure

Common Tools:

- Cisco Packet Tracer / GNS3
- Wireshark
- Draw.io (network diagrams)
- SNMP-based monitoring tools (e.g., PRTG, Zabbix)

Certifications:

- Cisco Certified Network Associate (CCNA)
- CompTIA Network+
- Cisco Certified Network Professional (CCNP) — advanced

Linux System Administrator

Responsibilities:

- Install, configure, and maintain Linux servers (e.g., Ubuntu Server)
- Manage users, permissions, services, and system security
- Automate routine tasks using shell scripting

- Perform backups, patching, and performance monitoring

Skills:

- Command-line proficiency (Bash) and shell scripting
- Understanding of file systems, services (systemd), and networking on Linux
- Package management (APT) and security hardening
- Familiarity with virtualization and containerization

Common Tools:

- SSH / Terminal
- Bash / Shell scripting
- Cron for task scheduling
- VirtualBox / Docker for virtualization

Certifications:

- CompTIA Linux+
- Red Hat Certified System Administrator (RHCSA)
- Linux Professional Institute Certification (LPIC-1)

Cloud Administrator

Responsibilities:

- Deploy, monitor, and manage cloud-hosted resources (compute, storage, networking)
- Manage cloud backups, scaling, and disaster-recovery configurations
- Control cloud access, identity, and cost optimization
- Support the company's future migration or hybrid cloud strategy

Skills:

- Familiarity with a major cloud platform (AWS, Azure, or Google Cloud)
- Understanding of Infrastructure as Code (IaC) concepts
- Cloud security and identity/access management (IAM)
- Cost monitoring and resource optimization

Common Tools:

- AWS Management Console / Azure Portal
- Terraform / CloudFormation (IaC)
- Cloud monitoring dashboards (CloudWatch, Azure Monitor)
- Cloud backup services

Certifications:

- AWS Certified Solutions Architect – Associate
- Microsoft Certified: Azure Administrator Associate
- Google Associate Cloud Engineer

How These Four Roles Work Together

Inside Vantix Technologies, these four roles function as one connected support chain rather than isolated jobs. The Helpdesk Technician acts as the first point of contact for everyday employee issues, resolving simple problems quickly and escalating anything deeper. When an escalated issue involves connectivity, the Network Administrator steps in to diagnose routers, switches, the firewall, or wireless coverage, since a stable network is the foundation everything else depends on. The Linux System Administrator keeps the on-premise Ubuntu Server, internal applications, and Git repository running securely, and works closely with the Network Administrator whenever server access or firewall rules need to be adjusted. As the company grows and begins to rely on cloud-hosted backups or services, the Cloud Administrator extends this same responsibility beyond the office walls, ensuring that on-premise and cloud resources stay in sync, secure, and cost-efficient. Together, these four professionals form a layered support model — endpoint, network, server, and cloud — that allows a small IT team of five to reliably support a twenty-person startup.

Part 7 — Infrastructure Recommendations

Internet Provider

A fiber-optic business internet plan (e.g., PLDT Business Fibr or Converge ICT Business) with a minimum of 100 Mbps symmetrical bandwidth and a Service Level Agreement (SLA) guaranteeing uptime and priority support is recommended. A business-grade line, rather than a residential plan, ensures a static IP (useful for remote access and hosting internal services), faster restoration times, and support for VPN and future cloud connectivity.

Server Specifications

A single rack/tower server is sufficient for a 20-person startup at launch: Intel Xeon E-2300 series or AMD Ryzen Pro CPU (6–8 cores), 32 GB ECC RAM, 2 x 1TB SSD in RAID 1 for redundancy, dual Gigabit NICs, and hot-swappable drive bays. Running Ubuntu Server 22.04 LTS with VirtualBox allows the server to host file sharing, the internal Git repository, and small internal web applications on isolated virtual machines, while ECC RAM and RAID 1 protect against data corruption and single-drive failure.

Backup Strategy

Implement the 3-2-1 backup rule: at least three copies of data, on two different media types, with one copy stored offsite. Concretely: (1) primary data on the server/NAS, (2) a nightly automated backup to the NAS storage device, and (3) a weekly rotation to the external backup drives, one of which is kept offsite (e.g., taken home by the IT lead or stored in a fireproof safe). As the company grows, a low-cost cloud backup service can be added as the true offsite copy.

Security Recommendations

Segment the network using VLANs to separate Finance/HR (sensitive data) from general staff and guest Wi-Fi traffic. Configure the firewall to block unused inbound ports, enable intrusion detection, and log traffic. Require multi-factor authentication (MFA) for email and any remote-access tools. Keep all systems patched through a monthly update schedule, and restrict administrative rights on end-user machines to IT staff only.

Antivirus

Microsoft Defender for Business (bundled with Microsoft 365 Business Standard/Premium) is recommended as the baseline endpoint protection, since it is centrally manageable, requires no separate licensing cost if already on Microsoft 365, and integrates with Windows 11 Pro. Real-time protection, scheduled scans, and centralized alerting should be enabled for every endpoint, including the server.

Password Policy

Enforce a minimum password length of 12 characters with a mix of upper/lowercase letters, numbers, and symbols; require password changes every 90 days for privileged (admin) accounts; disable password reuse for the last 5 passwords; and lock accounts after 5 failed login attempts. Passwords should never be shared, and a company-approved password manager should be provided for storing credentials securely.

Expansion Plan

The current design supports up to roughly 30 users before the switch and wireless access points need upgrading. As the company grows: (1) add a second managed switch and stack it with the first for more wired ports, (2) add a third wireless access point for full-floor coverage or additional office space, (3) migrate from a single server to a small virtualization cluster or hybrid cloud setup once workloads increase, and (4) revisit the internet plan's bandwidth tier once video conferencing and cloud usage grow. Planning IP address ranges and VLANs with extra headroom now avoids re-addressing the network later.

Part 8 — Personal Reflection

Working on the Enterprise Infrastructure Planning Project for Vantix Technologies gave me a much clearer picture of what a System Administrator actually does before a single cable is ever plugged in. Coming into this project, I assumed that "IT infrastructure" mostly meant buying computers and connecting them to the internet. By the time I finished mapping out the hardware, software, and network requirements for a 20-person company, I realized how much analysis, planning, and justification happens long before any purchase order is signed.

The most valuable thing I learned was how every decision in an infrastructure plan is connected to a business reason, not just a technical preference. For example, deciding how many laptops versus desktops to buy wasn't just about cost — it depended on which departments needed mobility (like Sales, who visit clients) versus which departments needed fixed, more secure workstations (like Finance, who handle sensitive financial data). This taught me that a good System Administrator has to understand the business itself, not just the technology.

The most challenging part of this project was designing the network diagram and making sure it was both technically accurate and easy to read. It was tempting to keep the diagram simple, but the requirements asked

for firewalls, switches, access points, and all four departments to be represented with proper connections. Balancing clarity with completeness took several revisions — I had to think about the diagram the way an IT Manager would read it, not just the way I understood it in my head. Writing realistic justifications for hardware and software quantities was similarly challenging, since I had to think through actual workflows (e.g., why the IT team needs a second monitor, or why Finance should not use shared laptops) rather than just filling in a table.

This project reinforced why planning matters so much before deployment. If Vantix Technologies bought equipment first and planned later, they could easily end up with the wrong number of devices, a network that can't scale, or security gaps that are expensive to fix after the fact. Planning forces you to anticipate problems — like power outages, data loss, or growth beyond 20 employees — while they are still cheap and easy to solve on paper. It also produces documentation that other IT staff, or even non-technical executives, can use to understand what was built and why.

Overall, this project helped me practice thinking like a System Administrator rather than just a computer user. I practiced translating business needs into a concrete hardware and software inventory, designing a secure and scalable network topology, and writing documentation that a real IT Manager could actually use. These are skills I expect to rely on constantly in future internships and jobs, since every organization — no matter how small — eventually needs someone who can plan its infrastructure instead of just reacting to problems as they happen.